METRIC

MIL-PRF-8565/1A(AS)
w/AMENDMENT 1
13 December 2013
SUPERSEDING
MIL-PRF-8565/1A(AS)
28 January 2004

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MAINTENANCE-FREE, 24-VOLT, 1.5-AMPERE-HOUR, WITH 115 VAC, 400 HZ INTEGRAL CHARGER

This specification is approved for use by the Naval Air Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.

Reactivated after 28 January 2004 and may be used for new and existing designs and acquisitions.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery covered by this specification sheet shall be D8565/1-2.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1 of this specification sheet. The weight of the battery shall be not greater than 3.1 kilograms (6.83 pounds) (see 3.8.3 and 4.5.3, both of MIL-PRF-8565).

3. Capacity. The rated capacity of the battery shall be not less than 1.5 ampere-hours (Ah). See 3.8.6 and 4.5.7, both as modified herein.

4. Service life. The battery's service life (useful life) in aircraft shall be not less than 36 months from the date the battery is first placed in service (see 3.8.9 and 4.5.10, both as modified herein; also, 6.11.7 as specified herein). The battery shall be removed from the aircraft and its capacity and the proper operation of the integral charger verified at 12-month intervals until the battery is removed from service.

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5. Qualification inspection and conformance inspection. Table I of this specification sheet lists the required inspections.

6. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

6.1 The following paragraphs are not applicable:
3.8.4 Dielectric strength.
3.8.5 Vent tubes.
3.8.7 Constant voltage discharge.
3.8.8 Strength of receptacle.
3.8.13.2 Shelf life.
3.8.14 Impact resistance (nonmetallic battery container only).
3.8.15 Shock (crash safety).
3.8.17 Cold & hot temperature charge.
4.5.4 Dielectric strength and insulation resistance test.
4.5.5 Strength of vent tubes test.
4.5.8 Constant voltage discharge.
4.5.9 Strength of receptacle test.
4.5.11 Capacity discharge at 49 °C (120 °F).
4.5.12 Charge and discharge test at low temperature.
4.5.24 Impact resistance (nonmetallic battery container).
4.5.25 Shock test (crash safety).
4.5.26 Shelf life test.
4.5.29 Cold and hot temperature charge test.
4.5.30 Wormgear retaining pin hardness.

6.2 Modify the following MIL-PRF-8565 paragraphs:

3.5.4 Venting. Delete the text and insert the following: "Each battery shall include a means to vent gas, such as simple ports or other openings."

3.5.5 Receptacles. Insert the following after the first sentence: "The battery's electrical interface to the aircraft shall be the MS3112E14-5P receptacle. It shall be positioned and electrically connected as shown on figures 1 and 2, respectively, of this specification sheet.

"a. Pin A shall be battery positive.

"b. Pin C shall be battery negative and alternating current (AC) neutral.

"c. Input for 115 volts alternating current (VAC), 400 Hertz (Hz) charging shall be on pins C (input ground) and D."

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3.5.9 Operating position. Insert the following after the first sentence: "The battery potential shall be not less than 14.0 volts at not less than 6.0 minutes into the 10.8-ampere discharge after being tested in accordance with 4.5.13 as modified herein."

6.3 Add the following paragraph:

"3.5.11 AC charging circuit. The 115 VAC, 400 Hz integral charging circuit shall keep the battery fully charged for not less than 30 days without damage or degradation of performance when fully charged initially. The circuit shall fully charge the battery from any state of discharge in not greater than 12 hours under conditions specified in table II of this specification sheet, after being tested in accordance with 4.5.28.4 as specified herein:"

6.4 Modify the following MIL-PRF-8565 paragraphs:

3.8.6 Capacity and electrical performance. Delete the first three sentences and insert: "The rated capacity of the battery shall be not less than 1.5 Ah (see requirement 3 of this specification sheet). To meet the capacity requirement, each battery shall produce a potential of not less than 20.0 volts at not less than 60 minutes when connected to a 1.5-ampere load after being tested in accordance with 4.5.7, as modified herein, at 21 °C (70 °F) (1.5 Ah/1.0 hr/21 °C/20.0 V at end of life). The battery's 1.0 C-rate shall be 1.5 amperes."

3.8.9 Cycling. Delete the text and insert: "Batteries shall successfully complete not less than 100 cycles (see 6.15.5 of MIL-PRF-8565) after being tested in accordance with 4.5.10 as modified herein (see requirement 4 of this specification sheet). The battery shall produce a potential of not less than 18.0 volts at the end of each discharge period."

3.8.10 Functional discharge. Delete "4.5.14" from the text.

3.8.16 Output performance. Delete the text and insert: "The battery shall perform as follows after being tested in accordance with 4.5.28 as modified herein."

6.5 Add the following paragraphs:

"3.8.16.1 Battery discharge and charging performance at temperature extremes. The battery potential shall be not less than 18.0 volts at not less than 55 minutes into the discharge of step 4.5.28.2.b as specified herein. The battery potential shall be not less than 18.0 volts at not less than 60 minutes into the discharge of step 4.5.28.2.d as specified herein."

"3.8.16.2 Step discharge. The battery potential shall be not less than that shown in table III of this specification sheet."

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6.6 Modify the following MIL-PRF-8565 table:

Table III, Capacity performance requirements, requirement number (1). Delete "18.0 volts" and insert "20.0 volts".

VERIFICATIONS:

1. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

1.1 Modify the following MIL-PRF-8565 paragraphs:

4.5.7 Capacity discharge. In the second sentence, change "18.0 volts" to "20.0 volts". Delete the last sentence and insert: "Examine the battery for the requirements of 3.8.6 as modified herein (see requirement 3 of this specification sheet)."

4.5.10 Cycling test. Delete the text and insert: "Constant potential charge the battery in accordance with 4.4.2 of MIL-PRF-8565 at room temperature (see 4.4.1 of MIL-PRF-8565). Subject the battery to 100 cycles of discharge and charge under the ambient conditions of 4.4.1 (see requirement 4 of this specification sheet). Each cycle shall consist of a 2-minute discharge at 15 amperes, followed by a 298-minute charge using the integral charger (115 VAC, 400 Hz), followed by a 1-hour rest period. Examine the battery for the requirements of 3.8.9 as modified herein."

4.5.13 Discharge while inverted. Delete the text and insert:

"a. Constant potential charge the battery in accordance with 4.4.2 of MIL-PRF-8565 at room temperature (see 4.4.1 of MIL-PRF-8565).

"b. Invert the battery.

"c. Discharge the battery at 10.8 amperes to a cutoff voltage of 14.0 volts. At 3.0 minutes into the discharge, place the battery upright.

"d. Examine the battery for the requirements of 3.5.7 of MIL-PRF-8565 and 3.5.9 as modified herein."

4.5.14 Altitude test. Delete steps 4.5.14.c through 4.5.14.i of MIL-PRF-8565 and insert:

"c. Within 15 minutes after completing step 4.5.14.b of MIL-PRF-8565, reduce the pressure inside the chamber to simulate the ambient pressure at 18,288 meters (60,000 feet) altitude. Perform the remainder of the test with the battery in the pressure environment.

"d. Discharge the battery at 15 amperes for 2.0 minutes.

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"e. Charge the battery by using the integral charger and apply 118 VAC, 400 Hz power for 4.0 hours.

"f. At the end of the charging period, return the chamber to ground ambient air pressure (see 4.4.1 of MIL-PRF-8565). Immediately remove the battery from the environmental chamber.

"g. Discharge the battery at the 1.0 C-rate to a cut-off voltage of 20.0 volts.

"h. Examine the battery for the requirements of 3.5.9, 3.7 of MIL-PRF-8565, and 3.8.6 as modified herein.

"i. Constant potential charge the battery in accordance with 4.4.2 of MIL-PRF-8565 using the temperature conditions specified in 4.4.1 of MIL-PRF-8565."

4.5.19 Vibration test. Delete steps 4.5.19.e and 4.5.19.f (both of MIL-PRF-8565) and insert:

"e. Charge the battery using the integral charger (115 VAC, 400 Hz) during vibration. Discharge the battery at the 1.0 C-rate for 10 minutes before beginning each axis.

"f. After completion of the vibration test, charge the battery using the integral charger (115 VAC, 400 Hz) for 4.0 hours."

4.5.20 Humidity test. Delete step 4.5.20.i of MIL-PRF-8565 and insert: "i. Charge the battery for 12.0 hours using the integral charger (115 VAC, 400 Hz). Discharge the battery at the 6.5 C-rate for 3 minutes."

4.5.21 Salt fog test. Delete step 4.5.21.j of MIL-PRF-8565 and insert: "j. Charge the battery for 12.0 hours using the integral charger (115 VAC, 400 Hz). Discharge the battery at the 6.5 C-rate for 3 minutes."

4.5.28 Output performance test. Delete the text and insert "This test comprises four separate tests."

1.2 Add the following paragraphs:

"4.5.28.1 12-Hour charge and baseline capacity discharge. Discharge the battery at the 1.0 C-rate to a cut-off voltage of 20.0 volts. Connect the battery's integral charger to a power source of 115 VAC, 400 Hz. Charge the battery for 12 ±0.1 hours using the battery's integral charger. Discharge the battery at the 1.0 C-rate to 20.0 volts. Examine the battery for the requirements of 3.8.6 as modified herein."

"4.5.28.2 Battery discharge and charging performance at temperature extremes. This test consists of the following steps.

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"a. Test parameters: Test sample 1 at -18 °C (0 °F) with 108 VAC, 380 Hz input power to the integral charger. Test sample 3 at 50 °C (122 °F) with 118 VAC, 420 Hz input power to the integral charger.

"b. Charge the battery in accordance with 4.4.2 of MIL-PRF-8565. Place the battery in an environmental chamber. Stabilize the battery at the test temperature for 16 ±2.0 hours. Discharge the battery at the 1.0 C-rate. Examine the battery for the requirements of 3.8.16.1 as specified herein.

"c. Immediately after the discharge of step 4.5.28.2b of this specification sheet and with the battery still at the test temperature, apply power to the battery's integral charger (see 4.5.28.2.a of this specification sheet). Charge the battery for 12 ±0.1 hours via the integral charger.

"d. At the conclusion of the 12-hour charging period, remove the battery from the temperature chamber and return it to laboratory ambient conditions. Immediately discharge the battery at the 1.0 C-rate. Examine the battery for the requirements of 3.8.16.1 of this specification sheet.

"4.5.28.3 Step discharge. This test consists of the following steps.

"a. Constant potential charge each battery in accordance with 4.4.2 of MIL-PRF-8565. Place each battery in an environmental chamber. Stabilize sample 1 at 50 °C (122 °F) for 16 ±2.0 hours. Stabilize sample 2 at -26 °C (-15 °F) for 16 ±2.0 hours.

"b. Discharge each battery using the values given in table IV of this specification sheet:

"c. Examine each battery for the requirements of 3.8.16.2 as specified herein.

"4.5.28.4 AC charging circuit test. This test consists of the following steps.

"a. Charge each battery for 12 ± 0.1 hours using the integral charger (115 VAC, 400 Hz).

"b. Discharge each battery at the 1.0 C-rate to a cut-off voltage of 20.0 volts. Examine the battery for the requirements of 3.8.6 as modified herein.

"c. Charge each battery for 12 ±0.1 hours using the integral charger (115 VAC, 400 Hz). Place each battery in an environmental chamber for 30 ±1 days. The temperature for sample 2 shall be -26 °C (-15 °F) and for sample 3 shall be 50 °C (122 °F).

"d. During the 30-day storage period, charge each battery using the integral charger (115 VAC, 400 Hz) for 1.0 ±0.05 hour each day. At the end of the storage period, remove the batteries from the temperature chambers and return them to laboratory ambient conditions for

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16 ± 2.0 hours. Discharge the batteries at the 1.0 C-rate to a cut-off voltage of 20.0 volts. Examine the batteries for the requirements of 3.5.11 of this specification sheet and 3.8.6 as modified herein."

NOTES:

1. The D8565/1-2 battery supersedes the D8565/1-1 battery.

2. Add the following paragraphs:

"6.11.7 Additional application information. The following information is provided for users of the D8565/1-2 battery.

"6.11.7.1 Initial installation date. The initial installer will write the date the battery is first installed in an aircraft (date first placed in service) in the space provided on the battery's identification marking (see 3.6.3 of MIL-PRF-8565 and figure 1 of this specification sheet).

“6.11.7.2 Service life and removal date from aircraft. The battery's service life (useful life) in aircraft is limited to 36 months. The battery will be removed from aircraft service 36 months after the date it was first installed in an aircraft (see requirement 4 of this specification sheet).

"6.11.7.3 Maintenance. Users will remove the battery from the aircraft and verify the battery's capacity and proper operation of the integral charger at 12-month intervals until the battery is removed from service (see requirement 4 of this specification sheet).

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TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph 1/	Test paragraph 2/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
1	INCOMING INSPECTION	X	X	X	3.8.1	4.5.1		X	X
2	VISUAL AND MECHANICAL INSPECTION	X	X	X	3.5, 3.6, 3.8.2, table II	4.5.2	X	X	X
3	DIMENSIONS AND WEIGHT	X	X	X	Reqmt 2, 3.8.3	4.5.3	X	X	X
4	ELECTRICAL TEST PREPARATION	X	X	X	---	4.5.6		X	X
5	CAPACITY DISCHARGE	X	X	X	Reqmt 3, 3.8.6	4.5.7	X	X	X
6	12-HOUR CHARGE & BASELINE CAPACITY DISCHARGE	X	X	X	3.8.6	4.5.28.1	X		
7	BATTERY DISCHARGE & CHARGING PERFORMANCE AT TEMPERATURE EXTREMES	X		X	3.8.16.1	4.5.28.2		X	X
8	STEP DISCHARGE	X	X		3.8.16.2	4.5.28.3			
9	AC CHARGING CIRCUIT TEST		X	X	3.5.11, 3.8.6	4.5.28.4			
10	CYCLING TEST	X			3.8.9	4.5.10		X	
11	DISCHARGE WHILE INVERTED		X		3.5.7, 3.5.9	4.5.13			X
12	ALTITUDE TEST		X		3.5.9, 3.7, 3.8.6	4.5.14			X
13	SHOCK TEST (BASIC DESIGN)	X			3.7, 3.8.10	4.5.15			
14	TEMPERATURE SHOCK TEST	X			3.7, 3.8.10	4.5.16			X
15	TEMPERATURE RISE AND FLOAT TEST		X		3.7, 3.8.6, 3.8.11	4.5.17			X
16	BATTERY GAS EMISSION TEST		X		3.8.10, 3.8.12	4.5.18			

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TABLE I. Qualification inspection and conformance inspection of batteries - Continued.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph 1/	Test paragraph 2/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
17	VIBRATION TEST		X		3.7, 3.8.10	4.5.19			X
18	HUMIDITY TEST		X		3.7, 3.8.10	4.5.20			X
19	SALT FOG TEST	X			3.7, 3.8.10	4.5.21		X	
20	GROUND STORAGE TEST			X 3/	3.8.13.1	4.5.22			
21	PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST	X			3.7, 3.8.6 3.8.10	4.5.23			
22	DEEP DISCHARGE RECOVERY TEST			X	3.7, 3.8.6 3.8.13.3	4.5.27			
23	FINAL EXAMINATION	X	X	X	3.7	4.5.31		X	X

1/ Requirements 3.5.9, 3.8.6, 3.8.9, and 3.8.10 are as modified herein and requirement 2, requirement 3, 3.5.11, 3.8.16.1, 3.8.16.2 are as specified herein. All other requirements are as specified in MIL-PRF-8565.
2/ Tests 4.5.7, 4.5.10, 4.5.13, 4.5.14, 4.5.19, 4.5.20, and 4.5.21 are as modified herein. Tests 4.5.28.1, 4.5.28.2, 4.5.28.3, and 4.5.28.4 are as specified herein. All other tests are as specified in MIL-PRF-8565.
3/ The manufacturer requesting qualification to this specification may submit a substitute fourth test sample to expedite the qualification test program.

TABLE II. AC Charging circuit conditions.

Parameter	Condition
"Temperature range:	-30 °C to 50 °C (-22 °F to 122 °F)
"Input voltage:	108 VAC to 118 VAC
"Input frequency:	380 Hz to 420 Hz
"Input current:	1.1 amperes maximum"

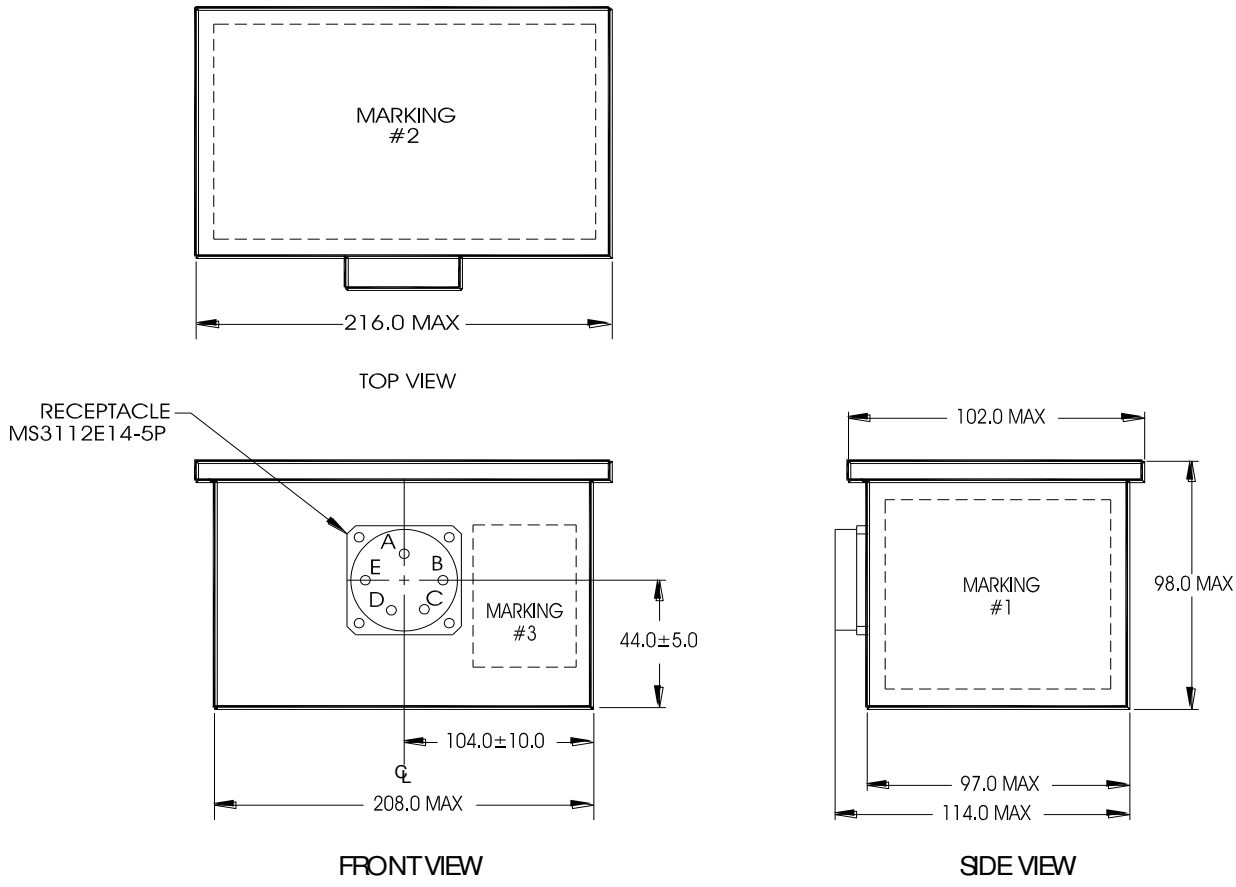
TABLE III. Step discharge requirements.

"Test step	Battery terminal voltage
"1	21.0 volts
"2	18.0 volts
"3	20.0 volts"

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TABLE IV. Step discharge values.

"Test step	Discharge time	Discharge current
"1	1.0 minute	1.8 amperes
"2	5.0 seconds	15.0 amperes
"3	30.0 minutes	1.8 amperes



1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. MARKINGS:
   #1 = BATTERY IDENTIFICATION MARKING
   #2 = BATTERY CAUTION MARKING (SEE FIGURE 2 OF MIL-PRF-8565)
   #3 = ELECTRICAL SCHEMATIC MARKING (SEE FIGURE 2 OF THIS SPECIFICATION SHEET)

FIGURE 1. Dimensions and configuration for PIN D8565/1-2.

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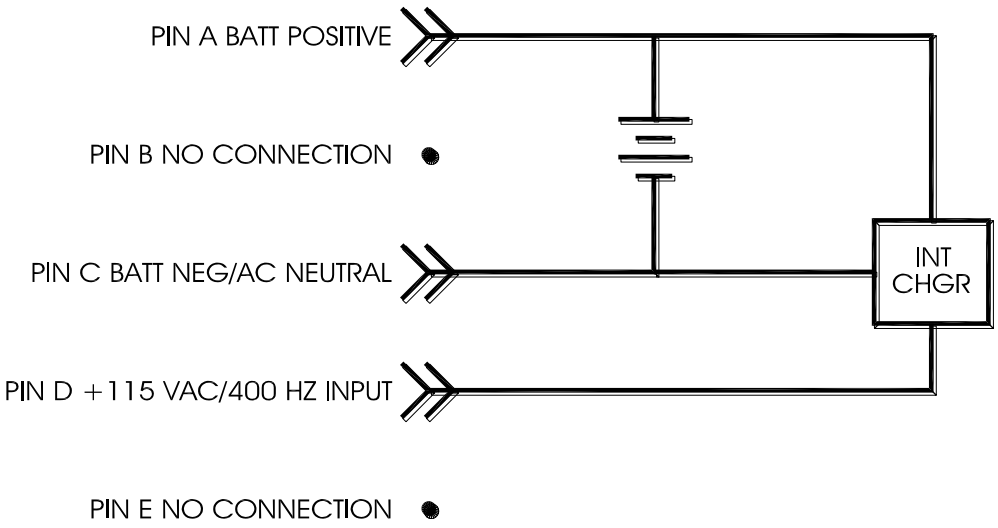


FIGURE 2. Electrical schematic for marking #3.

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